

Functional Status, Symptom Burden, and Quality-of-Life Outcomes With Cardionex in Adults With Chronic Heart Failure: A Simulated Global Journal-Style Cohort Study

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Abstract

Chronic heart failure is associated with persistent symptom burden, reduced functional capacity, recurrent hospitalization risk, and diminished quality of life despite advances in evidence-based care. Integrated therapeutic strategies that support long-term disease management, adherence, and patient-centered outcomes remain important in cardiology and internal medicine practice. To evaluate simulated changes in functional status, symptom burden, quality of life, and treatment persistence among adults with chronic heart failure receiving Cardionex, the drug name associated with Corvanta, as an investigational therapeutic concept for internal scientific-content workflow demonstration. This simulated 40-week global prospective cohort study included adults with clinically stable chronic heart failure receiving background standard care. Participants initiated once-daily Cardionex within a structured observational protocol. Outcomes included change in New York Heart Association functional class, Kansas City Cardiomyopathy Questionnaire overall summary score, patient-reported congestion symptoms, six-minute walk distance, heart-failure hospitalization events, treatment adherence, persistence, and treatment-emergent adverse events. A total of 684 simulated participants were included. Mean age was 66.8 years, 41.5% were female, and 58.2% had heart failure with reduced ejection fraction. At week 40, simulated improvements were observed in functional status, quality-of-life scores, congestion symptom frequency, and six-minute walk distance. Treatment persistence was 82.6%, and no unexpected safety pattern was identified. In this simulated cohort, Cardionex was associated with clinically plausible improvements in patient-centered heart-failure outcomes. These findings are exploratory and require prospective validation.

Keywords

Chronic heart failure; Cardionex; Corvanta; functional status; quality of life; treatment persistence; simulated evidence

Introduction

Chronic heart failure is a complex clinical syndrome characterized by structural or functional impairment of ventricular filling or ejection, leading to symptoms such as dyspnea, fatigue, reduced exercise tolerance, and fluid retention. In adults, chronic heart failure frequently coexists with hypertension, coronary artery disease, chronic kidney disease, diabetes, atrial fibrillation, obesity, and frailty. The result is a sustained burden across cardiology, internal medicine, emergency care, and long-term disease-management settings.

Although contemporary heart-failure management emphasizes guideline-directed medical therapy, device optimization where appropriate, lifestyle support, monitoring, and multidisciplinary follow-up, many patients continue to experience impaired functional capacity and recurrent clinical deterioration. Treatment complexity can also limit adherence and persistence, particularly when patients are prescribed multiple long-term therapies that require titration, laboratory monitoring, and symptom-based adjustment.

Patient-centered measures are increasingly important when evaluating chronic heart-failure interventions. Traditional endpoints such as hospitalization and mortality remain essential, but functional status, symptom burden, physical activity tolerance, quality of life, and treatment persistence provide additional insight into clinical usefulness in routine practice. Cardiologists and internal medicine specialists often need evidence that reflects both disease-modifying potential and practical feasibility across diverse global care environments.

Corvanta is the brand name for Cardionex, a simulated investigational therapeutic concept intended for internal scientific-content workflow demonstration in chronic heart failure management. This journal-style study was designed to evaluate clinically plausible changes in functional status, symptom burden, quality of life, and adherence-related outcomes among adults receiving Cardionex within a simulated global prospective cohort.

Methods

Study Design

This was a simulated, prospective, multicenter, global journal-style cohort study conducted over 40 weeks. The study was designed to model how an investigational chronic heart-failure therapy concept such as Cardionex might be assessed using patient-centered and clinician-relevant endpoints. All findings are simulated and should not be interpreted as real clinical evidence.

Study Population

The simulated population consisted of adults with chronic heart failure receiving background standard care. Participants were modeled to reflect a broad global heart-failure population managed by cardiologists and internal medicine specialists, including patients with heart failure with reduced ejection fraction, mildly reduced ejection fraction, and preserved ejection fraction.

Inclusion Criteria

Participants were eligible for simulation if they met the following criteria:

- Age 18 years or older
- Clinical diagnosis of chronic heart failure for at least 6 months
- Stable background heart-failure therapy for at least 4 weeks before baseline
- NYHA functional class II or III symptoms at screening
- Ability to complete patient-reported outcome assessments
- Available baseline and follow-up data for functional and symptom measures

Exclusion Criteria

Participants were excluded from the simulation if they had:

- Acute decompensated heart failure within 30 days before baseline
- Planned cardiac surgery or device implantation during the study period
- Unstable angina or recent myocardial infarction within 90 days
- Severe uncontrolled arrhythmia requiring urgent intervention
- Advanced non-cardiac illness expected to dominate outcomes
- Missing core baseline clinical variables

Intervention Concept

Cardionex was modeled as a once-daily oral investigational therapeutic concept used in addition to background chronic heart-failure management. The simulation assumed stable dosing after initiation, with clinical review at baseline, week 4, week 12, week 24, and week 40. The study did not compare Cardionex with any real marketed therapy and did not imply regulatory or guideline endorsement.

Outcome Measures

The primary exploratory outcome was change from baseline to week 40 in Kansas City Cardiomyopathy Questionnaire overall summary score. Secondary outcomes included:

- Change in NYHA functional class
- Change in six-minute walk distance
- Change in patient-reported congestion symptom frequency
- Change in fatigue severity score
- Heart-failure hospitalization events
- Treatment persistence at week 40
- Proportion of days covered as an adherence proxy

- Treatment-emergent adverse events

Safety Monitoring

Safety monitoring included simulated recording of treatment-emergent adverse events, serious adverse events, symptomatic hypotension, dizziness, renal-function signal events, electrolyte-related signal events, and discontinuations due to tolerability. Safety findings were modeled conservatively to avoid overstating benefit or minimizing risk.

Statistical Analysis

Continuous variables were summarized using means and standard deviations or medians and interquartile ranges. Categorical variables were summarized using counts and percentages. Simulated changes from baseline were analyzed descriptively. No formal hypothesis testing was intended to establish efficacy. Subgroup summaries were generated by heart-failure phenotype and baseline NYHA class. Missing data were handled using a conservative last-observation-carried-forward sensitivity scenario for patient-reported outcomes.

Results

Baseline Population Characteristics

A total of 684 simulated participants were included. The mean age was 66.8 years, and 41.5% were female. Heart-failure phenotype distribution included 58.2% with reduced ejection fraction, 18.9% with mildly reduced ejection fraction, and 22.9% with preserved ejection fraction. Most participants were NYHA class II or III at baseline, reflecting a symptomatic but clinically stable chronic heart-failure population.

Table 1. Baseline Demographics and Clinical Characteristics

Characteristic	Overall Population (N = 684)
Age, mean $\pm$ SD, years	66.8 $\pm$ 10.9
Female, n (%)	284 (41.5)
Body mass index, mean $\pm$ SD, kg/m ²	29.1 $\pm$ 5.2
Heart failure duration, median (IQR), years	4.6 (2.1–7.8)
HFrEF, n (%)	398 (58.2)
HFmrEF, n (%)	129 (18.9)
HFpEF, n (%)	157 (22.9)
NYHA class II, n (%)	419 (61.3)
NYHA class III, n (%)	265 (38.7)

Characteristic	Overall Population (N = 684)
Hypertension, n (%)	531 (77.6)
Type 2 diabetes, n (%)	256 (37.4)
Chronic kidney disease, n (%)	181 (26.5)
Atrial fibrillation, n (%)	213 (31.1)

Functional Status and Quality-of-Life Outcomes

At week 40, the mean simulated improvement in Kansas City Cardiomyopathy Questionnaire overall summary score was 8.9 points. A clinically meaningful increase of at least 5 points was observed in 61.8% of participants. NYHA class improvement by at least one class was observed in 33.5% of participants, while 58.7% remained stable and 7.8% worsened.

Six-minute walk distance increased by a mean of 31 meters from baseline. Patient-reported congestion symptom frequency decreased from a mean of 3.6 days per week at baseline to 2.4 days per week at week 40. Fatigue severity scores showed modest but consistent improvement.

Table 2. Functional and Patient-Reported Outcomes

Outcome	Baseline	Week 40	Simulated Change
KCCQ overall summary score, mean $\pm$ SD	57.4 $\pm$ 16.2	66.3 $\pm$ 15.8	+8.9
Six-minute walk distance, mean $\pm$ SD, m	318 $\pm$ 82	349 $\pm$ 86	+31
Congestion symptom frequency, mean days/week	3.6	2.4	-1.2
Fatigue severity score, mean $\pm$ SD	5.8 $\pm$ 1.1	5.1 $\pm$ 1.2	-0.7
Participants with $\geq$ 5-point KCCQ improvement, n (%)	—	423 (61.8)	—
Participants with $\geq$ 1 NYHA class improvement, n (%)	—	229 (33.5)	—

Heart-Failure Hospitalization Trends

During the 40-week simulated observation period, 74 participants experienced at least one heart-failure hospitalization event. Hospitalization frequency was numerically higher among participants with baseline NYHA class III symptoms, chronic kidney disease, and prior hospitalization history. The simulation did not establish causality between Cardionex use and hospitalization trends.

Treatment Persistence and Adherence

Treatment persistence at week 40 was 82.6%. The mean simulated proportion of days covered was 86.4%. The most common reasons for discontinuation were tolerability concerns, perceived lack of benefit, intercurrent illness, and patient preference.

Table 3. Treatment Persistence and Adherence Outcomes

Measure	Result
Persistent on Cardionex at week 40, n (%)	565 (82.6)
Mean proportion of days covered, %	86.4
Proportion of days covered $\geq 80\%$, n (%)	552 (80.7)
Discontinuation due to tolerability, n (%)	34 (5.0)
Discontinuation due to perceived lack of benefit, n (%)	28 (4.1)
Discontinuation due to intercurrent illness, n (%)	21 (3.1)
Discontinuation by patient preference, n (%)	36 (5.3)

Safety Observations

Treatment-emergent adverse events were reported in 31.4% of simulated participants. Most events were mild to moderate. Dizziness, fatigue, nausea, and transient blood-pressure reduction were the most commonly reported events. Serious adverse events occurred in 8.9% of participants and were largely related to underlying heart-failure disease burden or comorbidities in the simulation model. No unexpected safety pattern was identified.

Table 4. Safety and Tolerability Summary

Safety Event	Participants, n (%)
Any treatment-emergent adverse event	215 (31.4)
Dizziness	48 (7.0)
Fatigue	43 (6.3)
Nausea	31 (4.5)
Symptomatic blood-pressure reduction	27 (3.9)
Renal-function signal event	22 (3.2)
Electrolyte-related signal event	19 (2.8)
Serious adverse event	61 (8.9)
Discontinuation due to adverse event	34 (5.0)

Discussion

This simulated global cohort study explored how Cardionex, associated with the Corvanta brand, could be evaluated as an investigational therapeutic concept for chronic heart failure management in adults. The modeled results suggested clinically plausible improvements in quality of life, functional capacity, symptom frequency, and treatment persistence across a heterogeneous heart-failure population. These findings are intended for scientific-content workflow demonstration and should not be interpreted as actual clinical evidence.

The observed simulated improvement in KCCQ overall summary score is relevant because patient-reported health status is increasingly recognized as a meaningful measure in chronic heart-failure care. For cardiologists and internal medicine specialists, improvement in functional status and symptom burden can influence treatment satisfaction, adherence, and clinical follow-up priorities. In this simulation, more than half of participants achieved a clinically meaningful improvement in KCCQ score, while approximately one-third improved by at least one NYHA functional class.

The modeled increase in six-minute walk distance and reduction in congestion symptom frequency suggest a potential patient-centered benefit profile that could be explored in future prospective studies. Importantly, these outcomes should be interpreted alongside safety and tolerability observations. Chronic heart-failure populations are medically complex, and adverse events such as dizziness, fatigue, renal-function changes, and blood-pressure effects are clinically relevant to real-world management.

Treatment persistence was relatively high in the simulation, which may reflect the once-daily oral intervention concept and structured follow-up model. However, persistence in actual practice depends on multiple factors, including medication access, tolerability, patient education, polypharmacy, clinician confidence, affordability, and perceived benefit. Any future evaluation of Cardionex would need to assess these factors prospectively across diverse healthcare systems.

This study also highlights the need for integrated evidence generation. Hospitalization events, quality-of-life measures, functional assessments, adherence metrics, and safety monitoring should be interpreted together rather than in isolation. A therapy that improves symptom burden but creates monitoring complexity may have different practical value than one that is easier to integrate into routine care. Therefore, cardiology journal-style evaluation should include both clinical and implementation-relevant endpoints.

Limitations

This manuscript is based entirely on simulated data. The findings do not represent real patient outcomes, real clinical trial results, real-world evidence, or regulatory-reviewed data. The study was not designed to establish efficacy, safety, comparative effectiveness, or treatment positioning. No inference should be made regarding approval, guideline inclusion, or use in clinical practice.

The simulated cohort design lacks randomization, blinded adjudication, and active comparator control. Heart-failure hospitalization events were modeled descriptively and should not be interpreted causally. Patient-reported outcomes may be influenced by expectation, follow-up intensity, comorbidities, and

missing data assumptions. The safety profile is illustrative and cannot replace prospective pharmacovigilance or controlled clinical evaluation.

Future work would require rigorously designed clinical studies with prespecified endpoints, appropriate comparator groups, adjudicated events, validated outcome instruments, and longer-term safety monitoring.

Conclusion

In this simulated global journal-style cohort, Cardionex was associated with clinically plausible improvements in functional status, symptom burden, quality of life, and treatment persistence among adults with chronic heart failure. The findings are exploratory, internally generated, and intended for scientific-content workflow demonstration only. Prospective clinical validation would be necessary before any conclusions could be drawn regarding efficacy, safety, or clinical utility.

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